SEC PROJECT · FOR TODAY'S PRESENTATION

Presentation Script and Team Briefing

Design and Development of an Intelligent USB-Based Electronics Platform

WHO PRESENTS WHAT (about 7 minutes)

Slide	Topic	Speaker	Time
1	Title	Avi	0:45
2	Motivation and objectives	Avi	1:00
3	System architecture	Harshit	1:15
4	Instrument interface	Harshit	1:00
5	Software interface	Anshul	1:00
6	Proto and results	Anshul	1:15
7	Methodology and conclusion	Avi	1:00

THREE THINGS TO REMEMBER

- Avi opens by saying Soham could not be present. Keep it to one line and move on.
- Hand over clearly: "Now Harshit will explain the hardware." · "Now Anshul will explain the software and Proto." · "Avi will conclude."
- The graphs and the software screen show the **design and the test method with example data**. If asked, say so honestly, and state where the build actually stands.

1. The project in simple words

What it is: one small box that does the job of eight lab instruments. It connects to a laptop with one USB cable. One software screen controls everything, and an AI engineer called **Proto** runs the tests and explains the results in plain English.

Why: a normal lab bench needs many separate, costly instruments, each with its own controls. Students often get a reading but do not know what it means or what to fix.

How it works, in one breath: signal → protection and scaling (front end) → converter (analog to digital, or digital to analog) → FPGA, the fast "brain" chip → USB → laptop software → Proto explains.

THE EIGHT INSTRUMENTS, ONE LINE EACH

Instrument	What it does, simply
Oscilloscope	Shows how a voltage changes over time: the waveform
Waveform generator	Creates test signals, such as sine or square waves
Logic analyser	Watches digital 0/1 signals on 16 wires at once
Pattern generator	Sends chosen 0/1 patterns into a circuit
Protocol decoder	Turns digital "chatter" (I^2C, SPI, UART) into readable bytes
Power supply × 2	Powers the circuit at a set voltage and measures the current
Multimeter	Measures a voltage very precisely

Proto, the AI engineer: it follows a test plan, compares what was measured with what was expected, and says what is wrong and how to fix it.

2. Key parts and terms

Term	Meaning, simply
FPGA	A chip whose hardware we program. Very fast and works in parallel, so it can handle 65 million readings a second. We use a Tang Primer 20K.
ADC	Analog-to-digital converter: turns a voltage into numbers. Scope: AD9226. Multimeter: ADS1256.
DAC	Digital-to-analog converter: turns numbers into a voltage. Signal generator: AN9767.
12-bit	4,096 levels of resolution. 14-bit = 16,384 levels. 24-bit ≈ 16.7 million levels (very fine).
65 MS/s	65 million samples (readings) per second
Front end	Circuit before the ADC that protects and scales the input (AD8066 buffers, relays for ranges)
74LVC245	Buffer chip that protects and level-shifts the 16 logic lines
MCP4728 / INA226	MCP4728 sets the supply voltages; INA226 measures voltage and current
USB 2.0 High-Speed	Up to 480 Mbit/s link to the laptop; one cable for data
I²C, SPI, UART	Common ways chips talk to each other; the decoder shows these messages
Cutoff frequency (slide 6)	For an RC filter, $f = 1 / (2\pi RC)$. With R = 1 kΩ and C = 100 nF, f ≈ 1.59 kHz, where the gain drops by 3 dB
±5 % tolerance (slide 6)	A supply rail is accepted if it is within 5 % of its expected value

3. The script, slide by slide

Read it naturally; you do not need to memorise every word. Speak slowly and point at the slide while you talk.

Slide 1 · Title

AVI

about 0:45

Good morning. We are presenting our SEC project, Design and Development of an Intelligent USB-Based Electronics Platform, under the supervision of Prof. G.S. Chilana. I am Avi, with Harshit and Anshul; our team member Soham Pandit Nalawade could not be present today. In one line, our project is one small device that does the work of eight lab instruments, connects to a laptop with one USB cable, and has an AI engineer, called Proto, that runs tests and explains the results in plain language.

Slide 2 · Motivation and objectives

AVI

about 1:00

Why did we build this? A normal electronics bench needs five to eight separate instruments: an oscilloscope, a function generator, a logic analyser, a multimeter and power supplies. They are costly and bulky, each has its own controls, and students often see a number without knowing what it means or what to fix. So our objectives are: put all eight instruments in one USB unit; build the core on an FPGA; give everything one software screen; add Proto, an AI engineer that plans tests and explains results; and keep it affordable for student labs.

Slide 3 · System architecture

HARSHIT

about 1:15

This is how the hardware is organised. Read it from left to right. Each port on the left goes through a front end that protects and scales the signal, then through a converter. The two oscilloscope inputs use AD9226 chips, 12-bit converters that take 65 million readings per second. The waveform outputs use a 14-bit dual DAC. The 16 logic lines pass through a 74LVC245 buffer, the multimeter uses a precise 24-bit ADC, and the two power supplies are set by a small DAC and measured by an INA226 current monitor. Everything meets in the FPGA, the brain of the device. It handles triggering, captures data, generates waveforms and decodes buses, then sends everything over USB 2.0 to the computer, where our software and Proto run.

Slide 4 · Instrument interface

HARSHIT

about 1:00

This is the front panel the user actually touches. CH1 and CH2 are the oscilloscope inputs. W1 and W2 are the signal generator outputs. The 16-pin header is for logic signals: the logic analyser, the pattern generator and the protocol decoder all use it. V1, V2 and COM are the two programmable power supplies, DMM is the multimeter input, and USB-C connects to the laptop. The table shows each instrument, its port, its key specification and the chip that does the job.

Slide 5 · Software interface

ANSHUL

about 1:00

This is our software design. On the left you choose an instrument. In the middle is the oscilloscope screen with two waveforms, and below it the logic view decoding an I2C message into readable bytes. On the right are automatic measurements like frequency, voltage and rise time, and the Proto panel, where you can ask a question in plain English. On the right side of the slide is the software stack, from bottom to top: the FPGA logic, the USB link, the device interface, the workbench screen, and Proto at the top.

Slide 6 · Proto and results

ANSHUL

about 1:15

Proto is what makes the platform intelligent. It follows a fixed test flow: connect the board, power it up while watching the current, check every supply rail, drive the inputs, capture the outputs, decode the buses, give a verdict and suggest the fix. The first graph shows a frequency-response test of an RC filter. The generator sweeps the frequency, the scope measures the output, and Proto compares it with theory: the cutoff is one over two pi R C, about 1.59 kilohertz. The second graph checks each supply rail against a five percent tolerance. Here the sensor supply is 2.71 volts instead of 3.3, so Proto flags it and suggests what to check. These graphs show the test method with example data.

Slide 7 · Methodology and conclusion

AVI

about 1:00

> Our method has seven phases: requirements and architecture, component selection, schematic and PCB design, FPGA logic, the workbench software, Proto integration, and finally testing and validation. The expected outcome is one USB unit that replaces eight bench instruments, runs tests as plans, and explains results in simple language at a cost student labs can afford. In future we want to add a probe fixture that reaches every point on a board automatically, more protocol decoders, remote lab access and automatic lab reports. Thank you. We are happy to take your questions.

4. Likely questions and simple answers

Why an FPGA and not a microcontroller?

The scope takes 65 million readings a second from two channels at the same time. An FPGA handles this in parallel hardware with exact timing; a microcontroller is too slow for that.

Why USB?

One standard cable to any laptop. USB 2.0 High-Speed carries up to 480 Mbit/s, enough to stream the captured data.

What exactly is Proto?

An AI assistant inside the software. It follows a test plan, compares the measurements with the expected values, and explains in plain language what is wrong and what to try next.

What is the difference between the 12-bit and 24-bit ADCs?

The 12-bit ADC is very fast (65 million samples a second) for waveforms. The 24-bit ADC is slow but extremely precise, which suits the multimeter.

What is the highest frequency it can see?

In theory, half the sample rate (the Nyquist limit), about 32 MHz. In practice it is lower, because of the input circuit and the need for several samples per cycle.

How does the protocol decoder work?

The logic analyser records the 0/1 signals; the FPGA and software then recognise the protocol's rules (start, address, data, acknowledge) and show readable bytes.

How is it different from buying instruments separately?

Eight instruments in one box, one software screen, and Proto to explain results, at a much lower cost for a student lab.

Is it built? Are these real results?

Answer honestly with the current status. The graphs show the test method with example data.

Who did what?

Agree on this among yourselves before the presentation, including Soham's part.